

Opus Lugano Channel system



Producer statement cover sheet for structural glass balustrade systems

One cover sheet must be provided for each balustrade system

PART 1: TO BE COMPLETED BY THE DESIGN PROFESSIONAL / MANUFACTURER SUPPLYING THE GLASS SYSTEM

Cover sheet issued on behalf of:	Royal Glass
In respect of (system type):	Opus Lugano Channel System
Supplied to:	Auckland Council
Client / designers name:	
Description of work: <i>E.g. New dwelling</i>	
Address of property:	

PART 2: CONSTRUCTION DETAILS *(tick boxes that apply)*

Deck / balcony / stairs construction:	☐ Timber	☐ Precast	☐ Concrete	☐ Steel	☐ <i>please specify</i>	
Location of deck / balcony / stairs:	☐ Internal	☐ External	☐ <i>please specify</i>			
Deck / balcony / stairs:	☐ New	☐ Existing	Cladding:	☐ Direct fixed	☐ Cavity	☐ N/A

PART 3: GLASS DETAILS *(tick boxes that apply)*

Type:	☐ Toughened	☐ Toughened laminated	☐ Toughened laminated with stiff interlayer
Thickness:	(mm)	Width of pane:	0.5-1.8 (m)
Height above floor:	(m)	Height above fixing:	(m)

Note: If barrier extends more than 1.2m above fixing point, test reports must be provided

Supplier:	Royal Glass
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PART 4: BARRIER DETAILS

System name:	Opus Lugano Channel System
Drawings (ref page number)	
Engineers name:	Euro Glass
Hardware (type):	316 Stainless Steel

Drawings must show the type of hardware being used to support the glass (i.e. channel, discs, etc.)

PART 5: SAFETY FEATURES (tick all boxes that apply)

- | | |
|--------------------------|---|
| ☐ | Continuous interlinking rail (show how fixed to building and each pane of glass) |
| ☐ | Continuous interlinking top cap (show how fixed to building and each pane of glass) |
| ☐ | Clamps, brackets or point fixings (show how fixed to building and each pane of glass) |
| ☐ | Framed with structural sealant min. 2 edges (show how fixed to building and each pane of glass) |
| ☐ | Glass with stiff interlayer |
| ☐ | Two or three glass edges supported by continuous clamps |
| ☐ | Structural post to fix interlinking rail to deck |
| ☐ | Other (specify) |

Engineers name:

Euro Glass

Drawings (ref page number)

Comments:

PART 6: COVER SHEET COMPLETED BY

Name:

Roxy Huang

Signature:



Date:

Contact number:

0800 769 254

Email:

info@royalglass.co.nz

LUGANO BALUSTRADE SYSTEM RESIDENTIAL FACE FIX BALUSTRADE WITH 12MM GLASS

FOR RESIDENTIAL OCCUPANCY C3 OF TABLE 3.3 AS/NZS 1170.1

NOTES:

1) This proprietary balustrade system complies with New Zealand Building Code Clause B1 Structure, B1/AS1 Amendment 15, B1/VM1, F2 Hazardous Building Materials and F4 Safety From Falling Third Addition, subject to:

- all products meeting the required performance specification
- site installation carried out in accordance with intent of this drawing

2) Based on barrier loads from AS/NZS 1170.1, 12mm Toughened, 13.5mm Toughened Laminated Safety Glass, Maximum Height H above clamp are:

For residential occupancy C3 of Table 3.3 AS/NZS 1170.1

H: 1100mm high up to extra high wind zone of 2.13kPa

3) Design of support structure is the responsibility of others

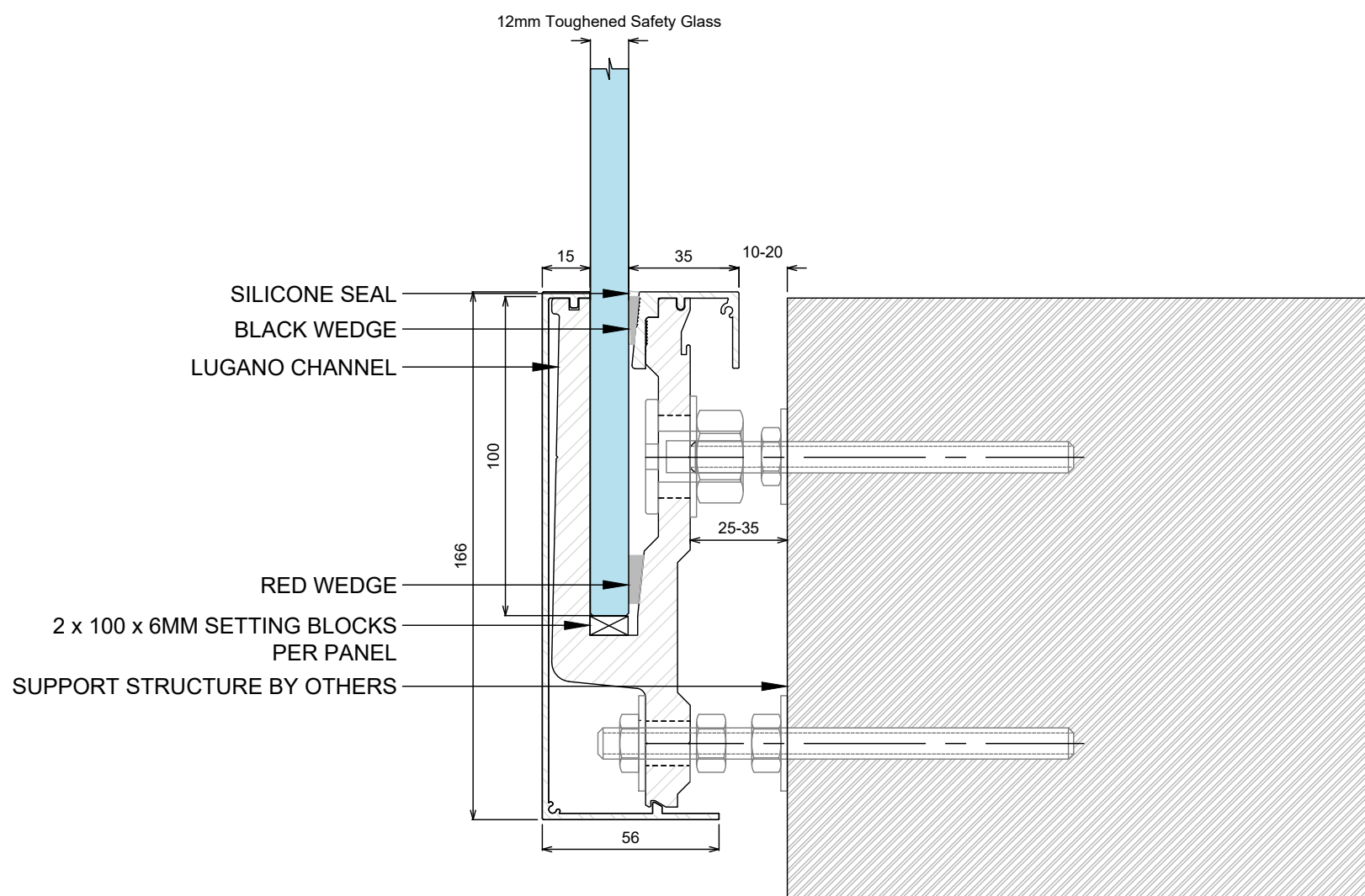
4) A handrail of 32-50mm diameter is required for stairs and ramps exceeding 1:20 slope. Refer NZBC D1/AS1

5) Safety glass Options according to 22.4.3 of NZS 4223.3:2016 are:

- 12mm Toughened Safety Glass with interlinking tail
- 13.2mm Toughened Laminated Glass with glass gap clamps and a maximum pane width of 2000mm
- 13.52mm Toughened SentryGlas Laminated Glass with minimum pane width of 1000mm

Interlinking rail or gap clamps must be connected to adjacent glass panes or to the building

6) The distance between fixings and the end of the channel or the joint must not exceed 300mm



THIS BALUSTRADE SYSTEM IS PROTECTED BY NZ PATENT 595645

LUGANO BALUSTRADE SYSTEM

RESIDENTIAL FACE FIX BALUSTRADE WITH 12MM GLASS

Side Fix LUGANO - Safety From Falling Barriers:

Concrete, steel & timber substrates only.

LAMINATED SENTRY toughened laminated safety glass, and TOUGHENED monolithic toughened safety glass.

Glass Thickness t (mm)	Occupancy	Maximum Design Height H (mm)	Channel Fixing Spacing (mm) Max	Design loads to deck structure		Maximum design wind pressure	
				M* (kNm/spacing)	T* (kN)	SLS Wind (kPa)	ULS Wind (kPa)
12, 13.2 & 13.52	C3	1100 (Concrete & Steel)	600	0.90	10.0	1.67 (Extra High)	2.13 (Extra High)
		1100 (Timber)	300	0.42	4.62	1.67 (Extra High)	2.13 (Extra High)

T* is tension force per fastener
(@ 300 or 600mm centres)

Side Fix LUGANO - Pool Fence (not protecting a fall > 1m):

Concrete, steel & timber only

Glass Thickness t (mm)	NZS 3604 Wind Zone	Substrate	Maximum Design Height H (mm)	Channel Fixing Spacing (mm) Max	Design loads to deck structure	
					M* (kNm/spacing)	T* (kN)
12, 13.2 & 13.52	Medium	Timber	1250 (13.52 = 1300)	300	0.42	4.61
		Concrete & Steel	1250 (13.52 = 1300)	600	0.83	9.30
	High	Timber	1250	300	0.46	4.61
		Concrete & Steel	1250	600	0.83	9.30
	Very High	Timber	1200	300	0.46	5.16
		Concrete & Steel	1200	600	0.93	10.32
	Extra High	Timber	1100* (see note)	300	0.46	5.16
		Concrete & Steel	1100* (see note)	600	0.93	10.32

Note: *Extra High wind zone requires a structural nib wall, fit to fence inside face. 1100+100mm nib = 1200mm

Glass thickness key: Note: Inner layer refers to balcony / walking surface side

Glass Thickness t (mm)	Inner layer glass thickness (mm)	Interlayer thickness (mm) and type	Outer layer glass thickness (mm)	Panel size requirements	
				Minimum panel width (mm)	Maximum panel width (mm)
12	-	-	-	1000	Refer manufacturing limits
13.2	6	1.2 LAMINATE	6	1000	Refer manufacturing limits
13.52	6	1.52 SENTRYGLAS	6	1000	Refer manufacturing limits

Maximum panel widths for Interlinking Rail/Bracket systems:

Applies where barrier is protecting a fall of 1.0m or more

Interlinking Rail System	Maximum panel width (mm)	Position
S25	1700	on glass only
S40	1700/1900	HB50 bracket / on glass
R40	1700/1900	HB50 bracket / on glass
AH40	1700/1900	HB50 bracket / on glass

Post failure requirements:

Applies where barrier is protecting a fall of 1.0m or more

TOUGHENED	Interlinking rail required in all cases
LAMINATED STF	No interlinking rail required Minimum panel widths apply

NOTES:

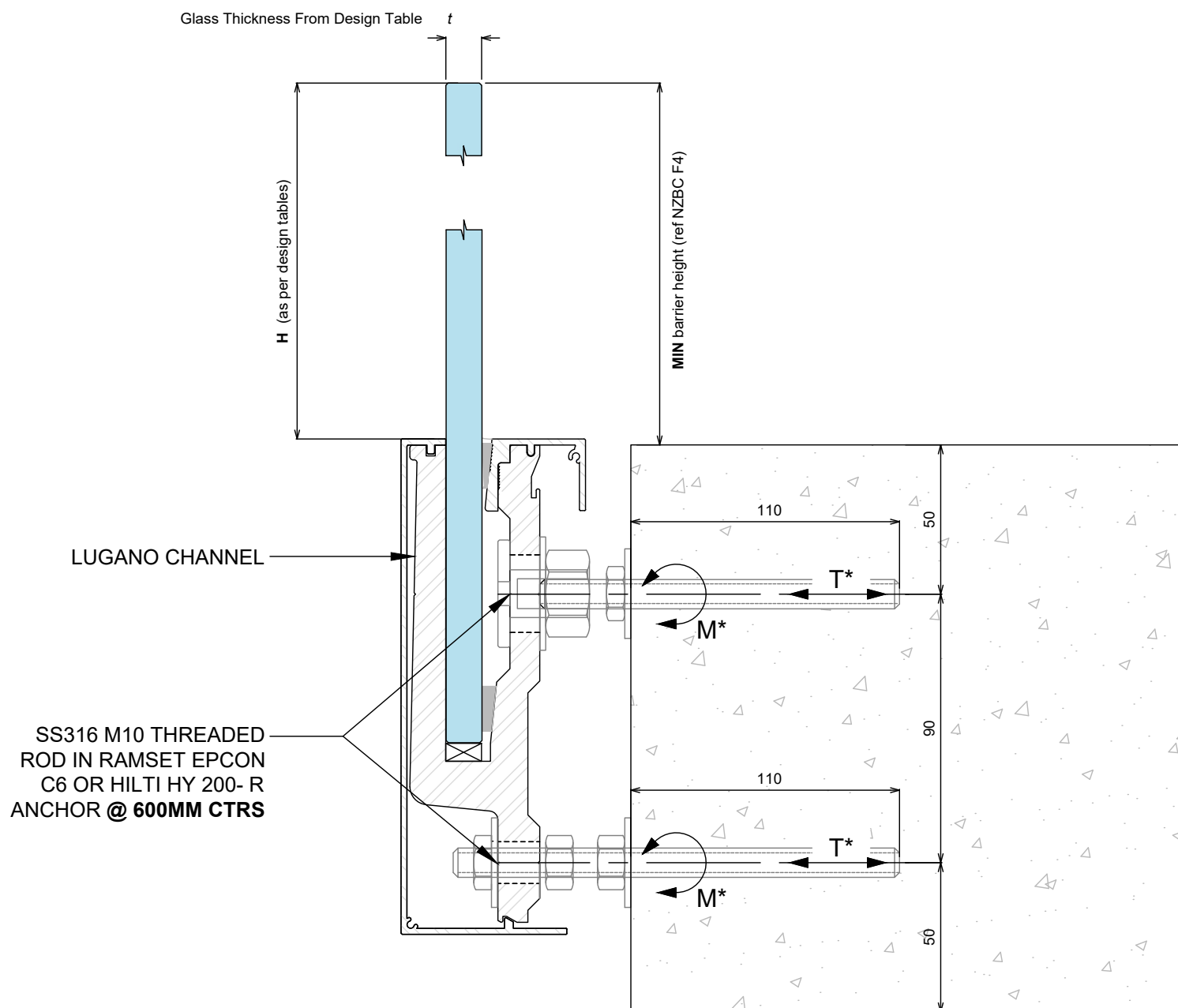
- Design table only valid for use with LUGANO balustrade system.
- Refer to elevation drawings for Height 'H'.
- The specifier must ensure the balustrade height above floor level requirements as per the NZ Building Code are complied with.
- Design loads are in accordance with AS/NZS 1170.1:2002 table 3.3 as modified by NZBC B1/VM1 and DBH Guidance on Barrier Design (March 2012).
- M* & T* denote bending moment (kNm/m width) and tension loads (kN/connection) respectively to be supported by the deck/pool structure.
- T* is based on wet SG8 timber substructures (except for occupancy A).
- Capacity of all substructure is to be verified by building engineer or checked for accordance with NZS3604 (where applicable) prior to fixing.
- All glass is to be toughened safety, in either TOUGHENED Monolithic or LAMINATED SENTRY variants subject to requirements of the tables above. Minimum glass residual strength 100MPa, all edges polished.
- Glass & interlayer thicknesses shown are nominal thickness. Table is based on glass minimum tolerance as per NZS 4223.1:2008.
- Refer to the relevant fixing details on drawings
- The tables for this balustrade system are based on an SLS deflection limit of 50mm. While greater than the suggested limit of height/60 as specified in NZS1170.0 for post and rail handrail systems, this is deemed acceptable based on the nature of the cantilevered glass system.
- For designs outside the scope of these tables and ULS wind pressures exceeding those shown, specific design is required.
- Maximum 10mm tolerance allowed to H heights noted in table.
- *Very High wind zone requires a structural nib wall, fit to fence inside face.
- LUGANO front cover does not possess toe hold.

LUGANO BALUSTRADE SYSTEM CONCRETE FIXING DETAIL

FOR RESIDENTIAL OCCUPANCY C3 OF TABLE 3.3 AS/NZS 1170.1

NOTES:

- 1) This proprietary balustrade system complies with New Zealand Building Code Clause B1 Structure, B1/AS1 Amendment 15, B1/VM1, F2 Hazardous Building Materials and F4 Safety From Falling Third Addition, subject to:
 - all products meeting the required performance specification
 - site installation carried out in accordance with intent of this drawing



NOTES:

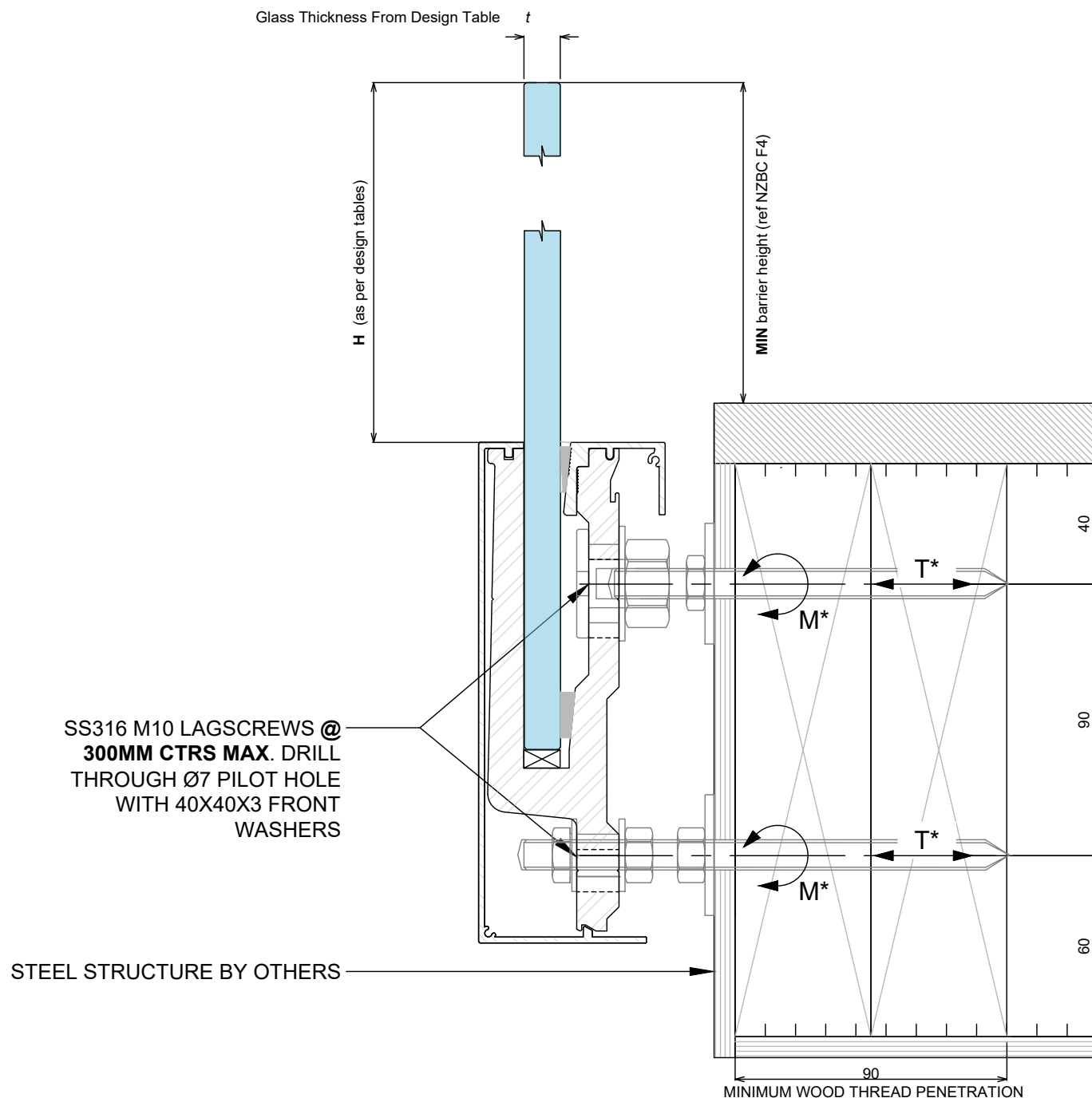
- 1) Capacity of structure is to be of sufficient strength to support loads M^* and T^* specified on balustrade system design table. Structure capacity to be verified by building engineer prior to fixing balustrade.
- 2) Max loading to comply with AS/NZS 1170.1:2002 Minimum Imposed Actions for Barriers Occupancy, shown at top of drawing, for design in accordance with balustrade system design table.
- 3) No substitution allowed - any variation from the details above and design tables will require specific design.
- 4) Substructure designer to ensure that there is no water runoff from dissimilar metals or treated timber onto barrier components.

LUGANO BALUSTRADE SYSTEM ENCLOSED DECK SG 8 TIMBER

FOR RESIDENTIAL OCCUPANCY C3 OF TABLE 3.3 AS/NZS 1170.1

NOTES:

- 1) This proprietary balustrade system complies with New Zealand Building Code Clause B1 Structure, B1/AS1 Amendment 15, B1/VM1, F2 Hazardous Building Materials and F4 Safety From Falling Third Addition, subject to:
 - all products meeting the required performance specification
 - site installation carried out in accordance with intent of this drawing



NOTES:

- 1) Capacity of structure is to be of sufficient strength to support loads M^* and T^* specified on balustrade system design table. Structure capacity to be verified by building engineer prior to fixing balustrade.
- 2) Max loading to comply with AS/NZS 1170.1:2002 Minimum Imposed Actions for Barriers Occupancy, shown at top of drawing, for design in accordance with balustrade system design table.
- 3) No substitution allowed - any variation from the details above and design tables will require specific design.
- 4) Substructure designer to ensure that there is no water runoff from dissimilar metals or treated timber onto barrier components.

LUGANO ELEVATION- RESIDENTIAL ONLY
OCCUPANCY C3

POOL BALUSTRADE

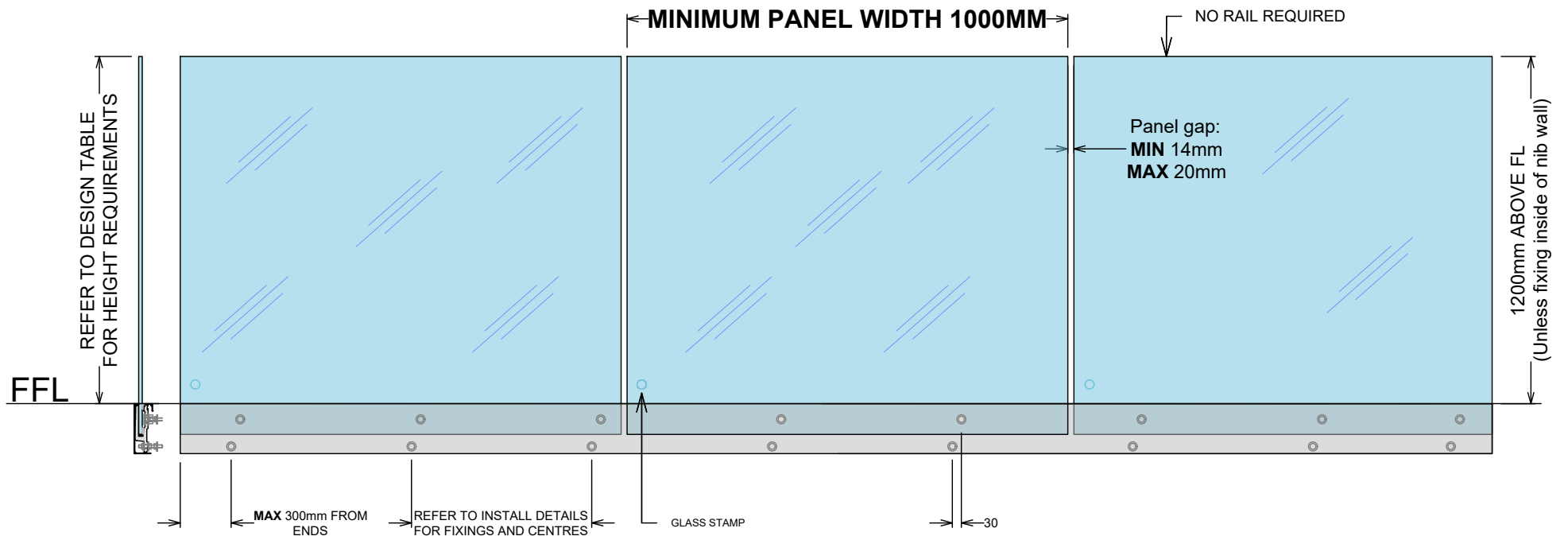
GLASS & FIXING SPECIFICATIONS:

Refer to design table for maximum glass height, maximum fixing spacing and design loads to structure.

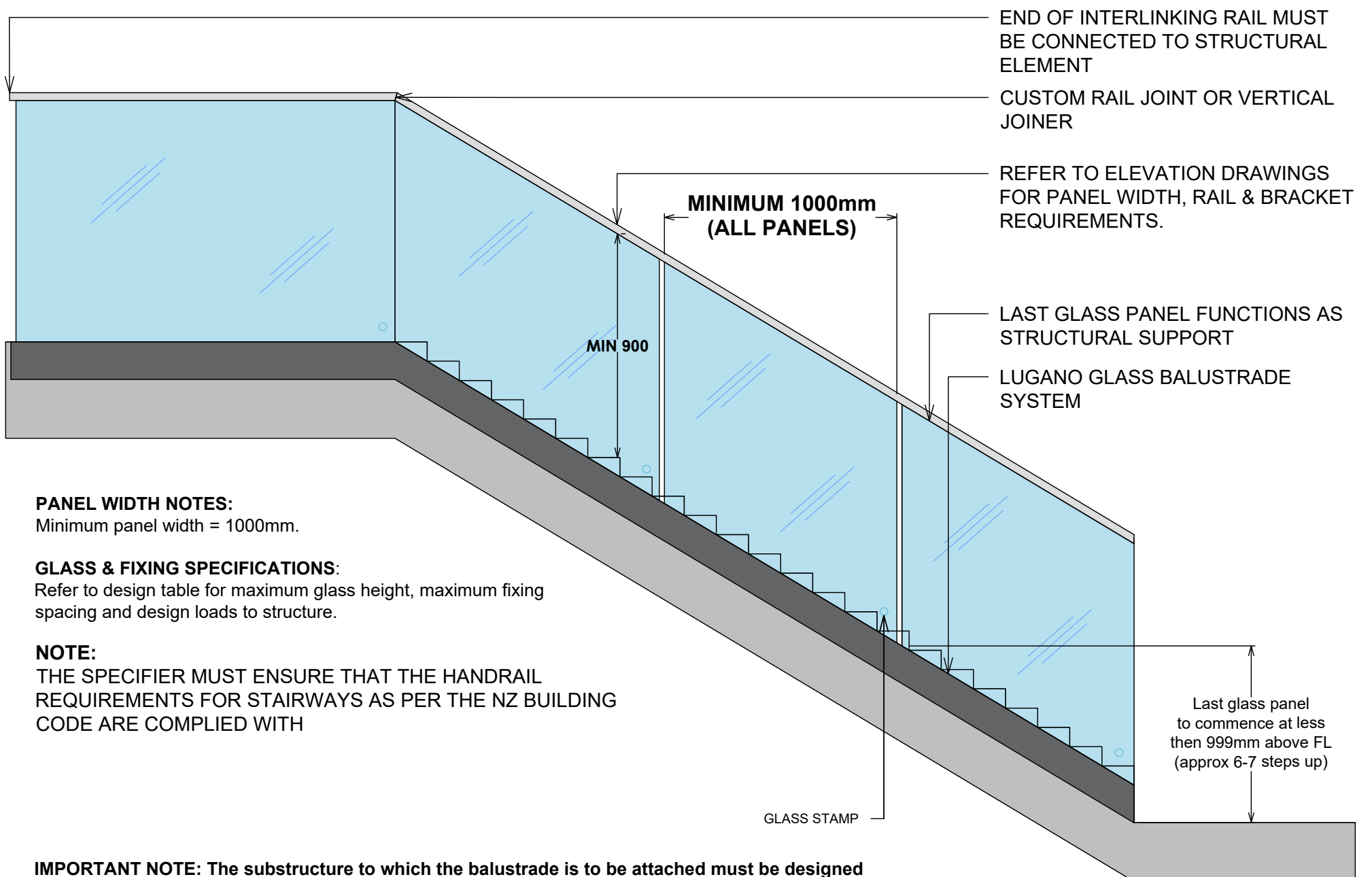
IMPORTANT NOTE: The substructure to which the balustrade is to be attached must be designed by a structural engineer to resist the relevant balustrade actions as per B1/VM1.

NOTES:

APPLIES TO FREE STANDING POOL FENCES NOT PROTECTING A FALL OF >1000mm
AS OF JAN 2017, COMPLIES WITH BUILDING CODE CLAUSE F9 & SECTION 162C OF THE BUILDING ACT



STAIR BALUSTRADE



PANEL WIDTH NOTES:

Minimum panel width = 1000mm.

GLASS & FIXING SPECIFICATIONS:

Refer to design table for maximum glass height, maximum fixing spacing and design loads to structure.

NOTE:

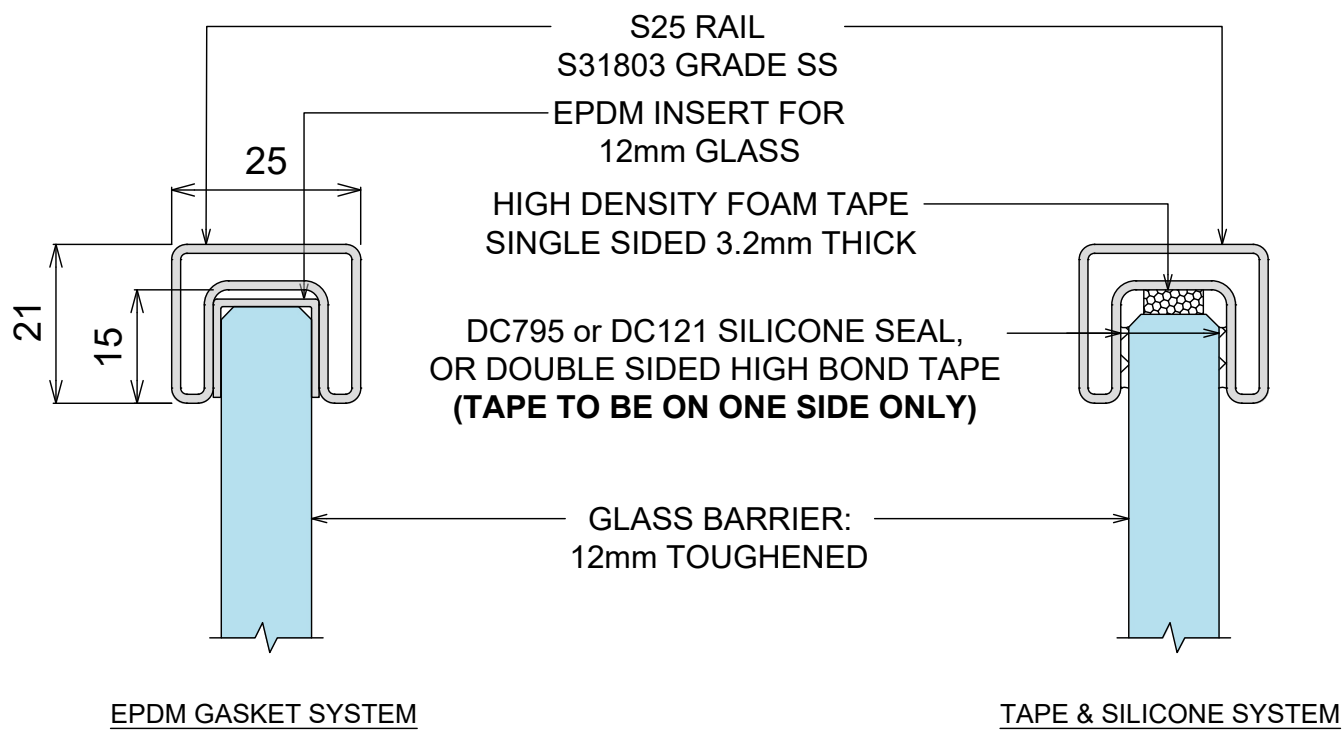
THE SPECIFIER MUST ENSURE THAT THE HANDRAIL REQUIREMENTS FOR STAIRWAYS AS PER THE NZ BUILDING CODE ARE COMPLIED WITH

IMPORTANT NOTE: The substructure to which the balustrade is to be attached must be designed by a structural engineer to resist the relevant balustrade actions as per B1/VM1.

LUGANO BALUSTRADE SYSTEM

S25 RAIL FOR 12mm TOUGHENED GLASS ONLY- OVERVIEW

REFER TO DESIGN TABLE & ELEVATIONS FOR LIMITATIONS & OPTIONS



COMPONENTS OVERVIEW

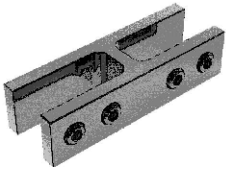
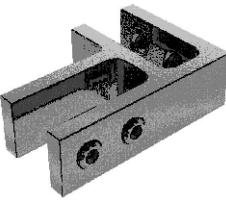
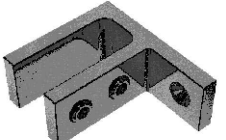
Ø5mm HOLE FOR M5 GRUB SCREW S25 STAINLESS RAIL	180° JOINER (& ADJUSTABLE VERTICAL JOINER)	90° JOINER (& ADJUSTABLE HORIZONTAL JOINER)	WALL BRACKET
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INSTALL OVERVIEW WALL BRACKET

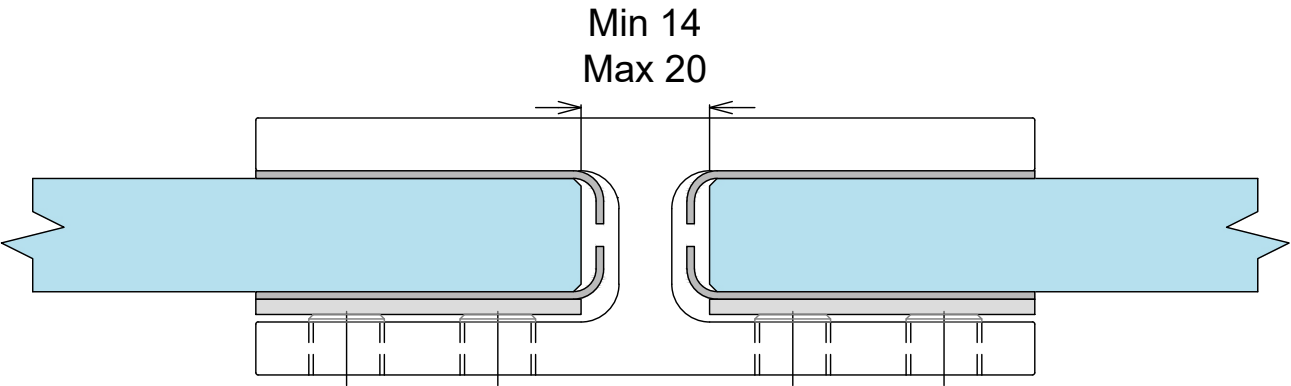
WALL BRACKET TO CONCRETE	WALL BRACKET TO BLOCKWALL	WALL BRACKET TO WEATHERBOARD	WALL BRACKET TO STEEL
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- NOTES:**
1. PREPARE FOR AND APPLY DC795 & DC121 STRUCTURAL SILICONE IN ACCORDANCE WITH DOWSIL PREPARATION AND INSTALLATION INSTRUCTIONS.
 2. INTERLINKING RAIL SPLICE & CORNER CONNECTIONS ARE SHOWN ON JOINER INSTALL OPTIONS DRAWING
 3. INTERLINKING RAIL END CONNECTION BRACKETS & ATTACHMENT DETAILS ARE SHOWN ON DRAWINGS WALL BRACKET INSTALL OPTIONS. ALL SCREWS TO BE STAINLESS STEEL WITH A MINIMUM ULTIMATE SHEAR STRENGTH OF 3.5kN (PER SCREW).
 4. LINK RAIL SECTION AND CONNECTION PIECES TO BE S31803 GRADE STAINLESS STEEL, IN ACCORDANCE WITH NZS 4673:2001.
 5. REFER TO WARRANTY & MAINTENANCE PAGES FOR PERIODIC INSPECTION, CLEANING & MAINTENANCE REQUIREMENTS.

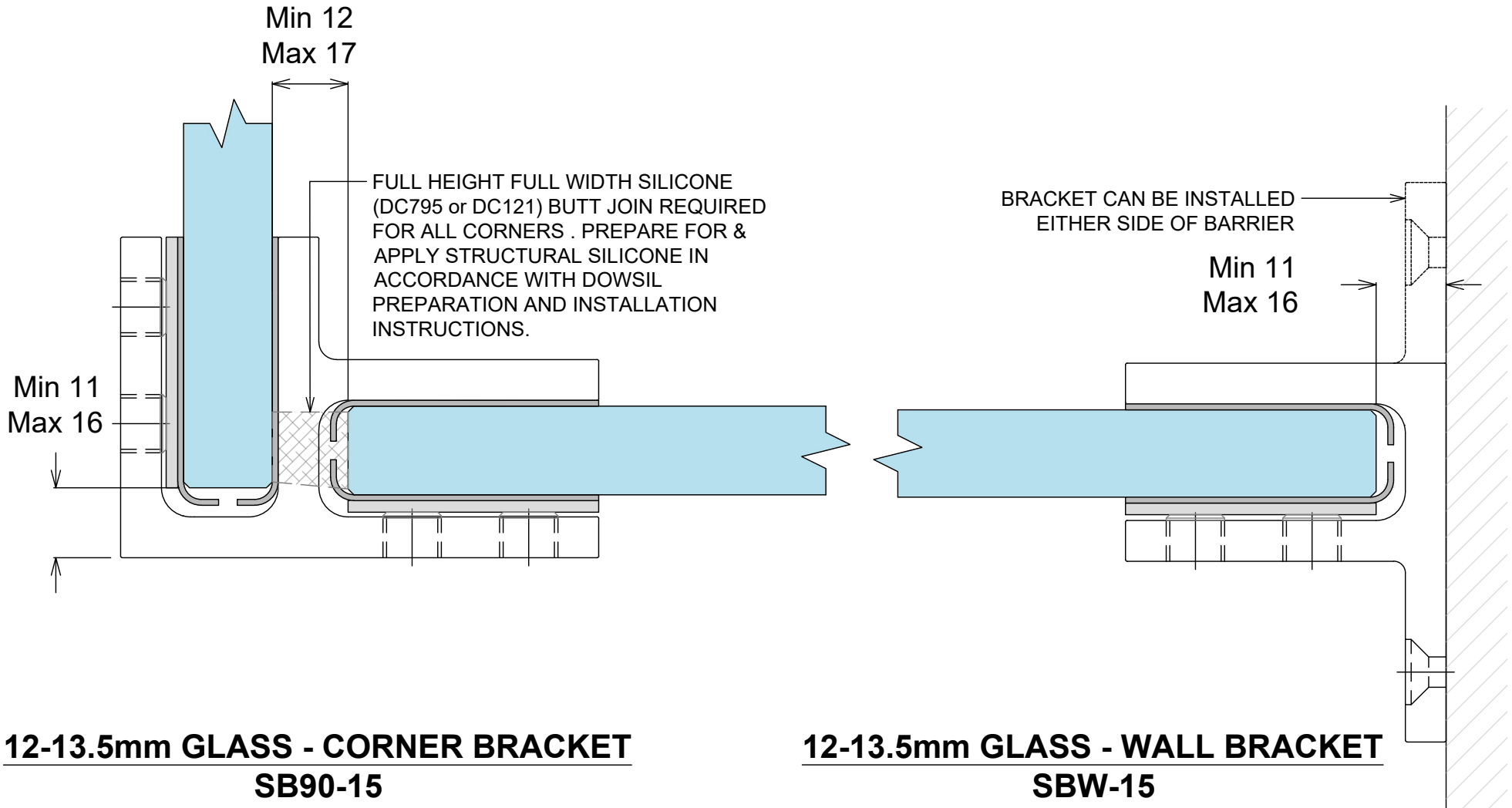
LUGANO BALUSTRADE SYSTEM
STIFFENER BRACKETS

Product	Code	Description
	SB180-15 (180° bracket)	Suit glass from 12 - 13.5mm. Stiffener bracket - No holes required. 180° glass to glass. Duplex 2205. Stainless Steel
	SB90-15 (90° bracket)	Suit glass from 12 - 13.5mm. Stiffener bracket - No holes required. 90° glass to glass. Duplex 2205. Stainless Steel
	SBW-15 (wall bracket)	Suit glass from 12 - 13.5mm. Stiffener bracket - No holes required. 90° glass to wall. Duplex 2205. Stainless Steel

All brackets are supplied with a selection of gaskets to suit glass thickness and includes stainless steel clamping plates.



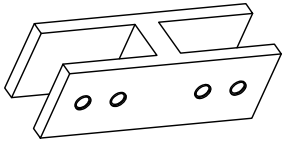
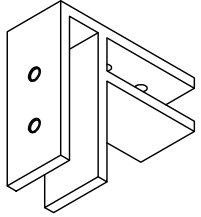
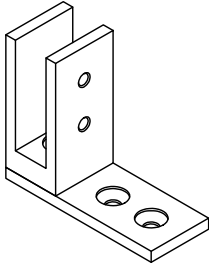
12-13.5mm GLASS - STRAIGHT BRACKET
SB180-15



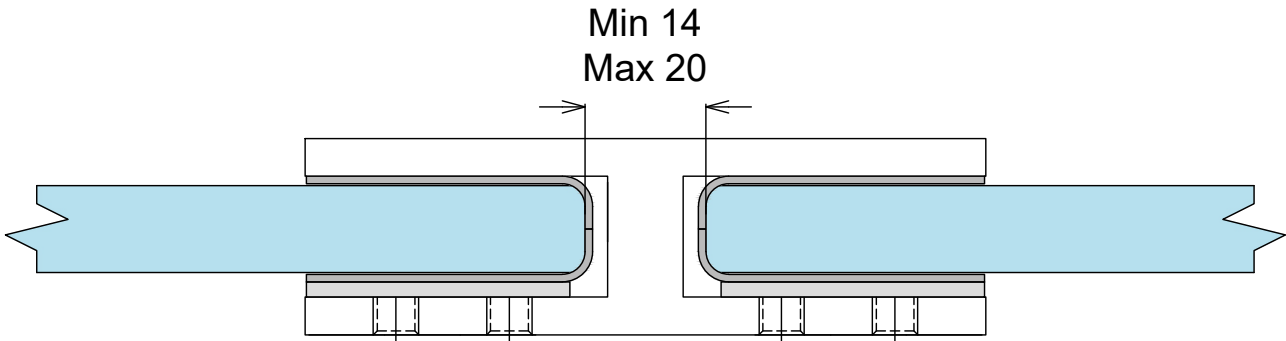
12-13.5mm GLASS - CORNER BRACKET
SB90-15

12-13.5mm GLASS - WALL BRACKET
SBW-15

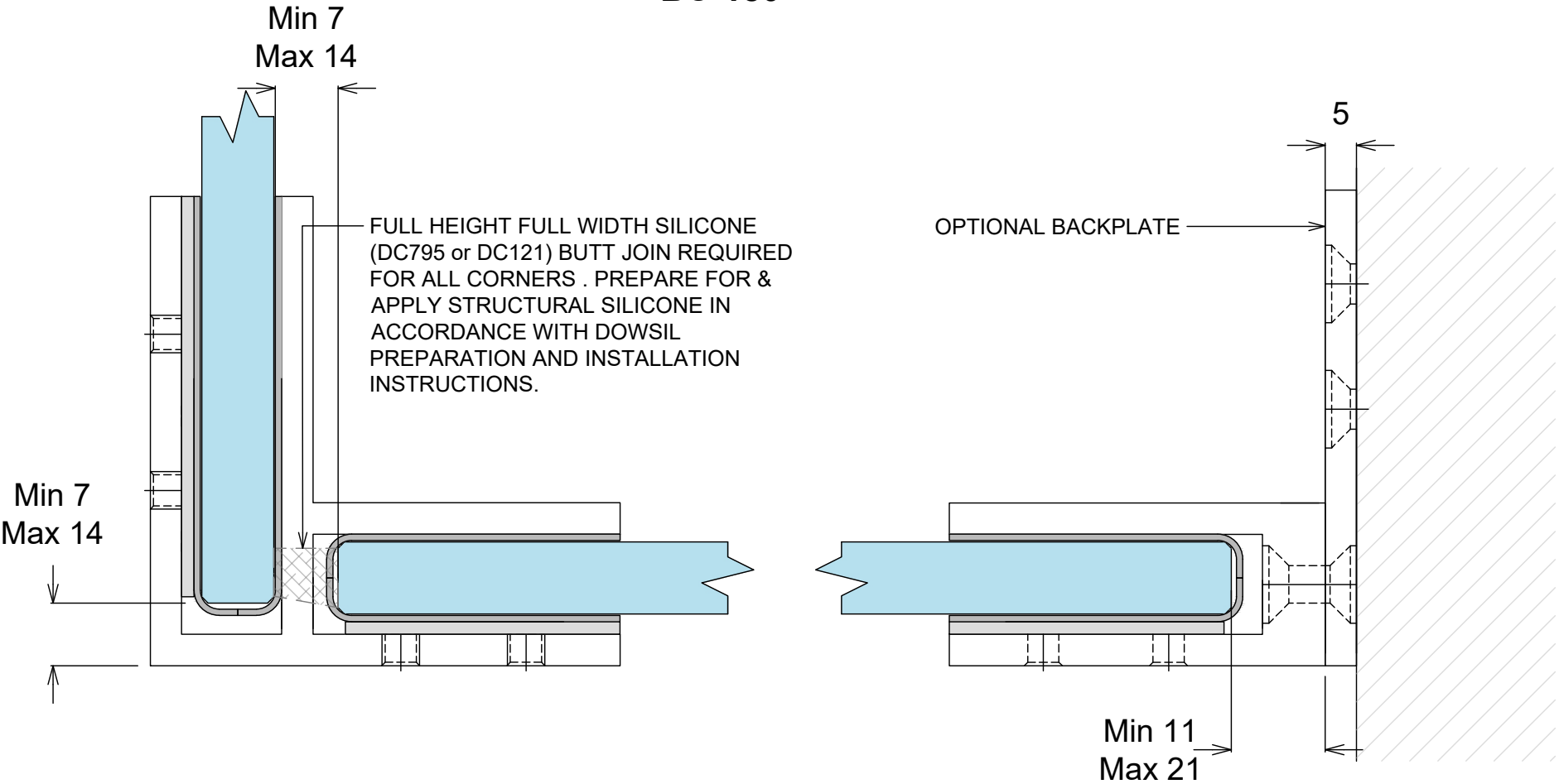
LUGANO BALUSTRADE SYSTEM
BALUSTRADE CLAMP

Product	Code	Description
	BC180 (180° bracket)	Suit glass from 12 - 13.5mm. 180° glass to glass. Duplex 2205. Stainless Steel Available In Polished, Satin & Black
	BCW (90° bracket)	Suit glass from 12 - 13.5mm. 90° glass to glass. Duplex 2205. Stainless Steel Available In Polished, Satin & Black
	BC-90 (wall bracket)	Suit glass from 12 - 13.5mm. 90° glass to wall. Duplex 2205. Stainless Steel Available In Polished, Satin & Black

All brackets are supplied with a selection of gaskets to suit glass thickness and includes stainless steel clamping plates.



12-13.5mm GLASS - STRAIGHT CLAMP
BC-180



12-13.5mm GLASS - CORNER BRACKET
BCW

12-13.5mm GLASS - WALL BRACKET
BC-90

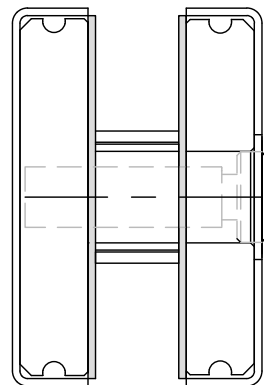
LUGANO BALUSTRADE SYSTEM
SLIMLINE HANDRAIL BRACKETS

ALL FIXINGS TO BE
STAINLESS STEEL

RT50H STAINLESS TUBE
50.8mm Ø HANDRAIL
SHOWN (ST50H, S40, R40
& AH40 ARE ALSO
COMPATIBLE)

SRS SLIMLINE ROUND
SADDLE (OPTION TO
WELD HANDRAIL TO
BRACKET)

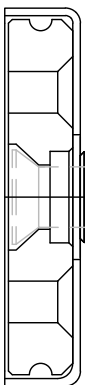
RSHBG ROUND SLIMLINE
GLASS MOUNT
HANDRAIL BRACKET
12MM



RT50H STAINLESS TUBE
50.8mm Ø HANDRAIL
SHOWN (ST50H, S40, R40
& AH40 ARE ALSO
COMPATIBLE)

SRS SLIMLINE ROUND
SADDLE (OPTION TO
WELD HANDRAIL TO
BRACKET)

RHSBW ROUND SLIMLINE
GLASS MOUNT
HANDRAIL BRACKET
12MM



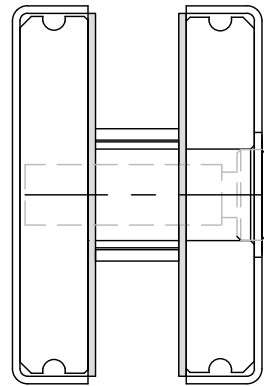
NOTE: AVAILABLE IN SATIN OR
POLISHED STAINLESS

ALL FIXINGS TO BE
STAINLESS STEEL

ST50H STAINLESS TUBE
50x25mm HANDRAIL
SHOWN (RT50H, S40, R40
& AH40 ARE ALSO
COMPATIBLE)

SFS SLIMLINE FLAT
SADDLE (OPTION TO
WELD HANDRAIL TO
BRACKET)

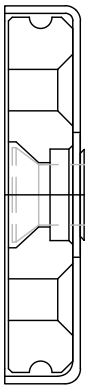
SSHBG SQUARE
SLIMLINE GLASS MOUNT
HANDRAIL BRACKET
12MM



ST50H STAINLESS TUBE
50x25mm HANDRAIL
SHOWN (RT50H, S40, R40
& AH40 ARE ALSO
COMPATIBLE)

SFS SLIMLINE FLAT
SADDLE (OPTION TO
WELD HANDRAIL TO
BRACKET)

SSHBW SQUARE
SLIMLINE WALL MOUNT
HANDRAIL BRACKET
12MM



NOTE: AVAILABLE IN SATIN OR
POLISHED STAINLESS

LUGANO BALUSTRADE SYSTEM

CARE AND MAINTENANCE

CARE AND MAINTENANCE OF BALUSTRADE GLASS & HARDWARE

Hardware requires regular maintenance to ensure the system performs at its best. As a general rule, the harsher the environment, the more regular the maintenance required to keep your hardware in top condition. Also hardware or systems that are covered by verandas or wide eaves and not subject to natural rain wash needs regular cleaning to avoid damage to surface finish on both the aluminium and any surface coated hardware.

Fair wear and tear;

- **Seals and rubbers** will require replacing from time to time depending on the environment. As a general rule, they should last for 10 years or more, and can be replaced by service provider.
- **Tracks, rollers** (if accessible) and hardware require lubrication; rollers may require replacing due to normal wear and tear. This depends on the environment and amount of use.

CARE AND MAINTENANCE, WASHING GLASS

Regular washing and drying of glass windows and doors are required to ensure their long term durability. In urban areas washing should be done every three to six months. The following guidelines apply:

- When washing, soak the glass surface with warm water and a mild soap detergent solution or proprietary glass cleaners to loosen dirt and debris.
- Use a soft grit free cloth or sponge when washing and try to avoid washing in direct sunlight. Do not use scrapers or razor blades.
- After washing, rinse with clean water and then dry the glass using a clean, grit-free squeegee, cloth, or paper towel. Remember, wet glass is dirty glass.
- All water and cleaning solution residue should be dried from the window gaskets, sealants, and frames to prevent water spots.
- Avoid cleaning tinted and reflective glass surfaces in direct sunlight. When washing special glass, the following guidelines apply:
- When washing double glazing and laminated glass, use the same procedures as above but ensure no solvents come into contact with the edge laminate interlayer or unit sealant.
- With reflective or Low E coated surfaces, exercise special care when cleaning - special cleaners may be required as they can be hard to clean. Follow manufacturer's instructions.
- It is advisable to check that frame drainage is not blocked this can affect laminate and insulated glass units.

DO NOT:

- Do Not Use Scrapers of any type or size on a Glass surface.
- Do Not - Leave building dirt or residues to remain on Glass for a period of time.
- Do Not - Begin cleaning glass until you have identified the surface type.
- Do Not - Clean Glass surfaces in direct sunlight.
- Do Not - Allow dirty water or cleaning residues to remain on the Glass.
- Do Not - Begin cleaning before rinsing off loose residues.
- Do Not - Use abrasive cleaning solutions, materials or solvents.
- Do Not - Allow metal parts of the cleaning equipment to come in contact with the Glass.
- Do Not - Trap abrasive particles between the cleaning material and the Glass.

DO:

- Clean glass promptly when dirt or building residues appear.
- Determine glass surface type.
- Exercise special care when cleaning coated surfaces.
- Avoid cleaning glass surfaces in direct sunlight.
- Start cleaning at the top of a building, then continue to lower levels.
- Soak the glass surface in a clean soapy solution before cleaning.
- Use a mild non abrasive commercial cleaner.
- Use a squeegee to remove all cleaning solution.
- Try your procedures on a small window and check.

LUGANO BALUSTRADE SYSTEM

CARE AND MAINTENANCE

CARE AND MAINTENANCE OF POWDERCOATING

Powder coating is available in a wide range of colours with commercially available surface integrity warranties from 10 to 30 years. The powder coating surface warranty period is conditional upon the formulation and micron thickness. Over time with exposure to the elements, powder coatings may show signs of weathering such as loss of gloss, chalking and slight color change. A simple regular clean will minimise the effects of weathering and will remove dirt, grime and other build-up detrimental to all powder coatings.

The frequency of such cleaning will depend on many factors including the:

- Geographical location of the building.
- Environment surrounding the building e.g. marine, industrial, alkaline or acidic, etc.
- Levels of atmospheric pollution including salts.
- Prevailing winds and the possibility of air borne debris causing erosive wear of the coating e.g. sand causing abrasion.
- Protection of part or all of the building by other buildings.
- Change in environmental circumstances during the lifetime of the building e.g. if rural became industrial.

The following guidelines apply:

- Just a gentle clean with a soft brush and mild detergent, followed by a fresh water rinse, will maintain the long-term performance of the powder coated or anodised aluminium. In rural or normal urban environments cleaning should occur every six months. In areas of high pollution, such as industrial areas, geothermal areas or coastal environments, cleaning should occur every three months. In particularly hazardous locations, such as beachfronts, severe marine environments or areas of high industrial pollution, cleaning should be increased to monthly.
- Sheltered areas can be at more risk of coating degradation than exposed areas. This is because wind-blown salt and other pollutants may adhere to the surface. These areas should be inspected and cleaned if necessary on a more regular basis.
- Adequate on site protection of delivered and/or installed hardware must be provided. Hardware may get knocked, scratched, or splattered with mortar, plaster, or paint during the later stages of construction. If splashes occur immediately wash down the hardware unit affected with water or methylated spirits* (*wash area thoroughly afterwards). Do not allow splashes to harden.
- To restore powder coated surfaces that have lost gloss or are chalking, polishing with a high quality crème polish in accordance with the manufacturer's instructions is recommended. Avoid polishes that contain cutting compounds, unless the surface is extremely weathered.

DO NOT USE SOLVENTS Strong solvent type cleaners should not be used. These are harmful to the extended life of your hardware system.

CARE AND MAINTENANCE OF ANODISING

Anodised hardware is not only attractive, but also offers a durable and tough wearing finish. Some deterioration of the anodic oxide coating may occur, mainly as a result of grime deposition and subsequent attack by moisture, particularly if the moisture is contaminated with sulphur compounds.

Regular cleaning is essential to preserve the finish of anodised aluminium over a long period. The following guidelines apply:

- Anodised aluminium should be washed with warm water and a suitable wetting agent or mild soap solution, in a similar manner to washing a car. A fine brush may be used to loosen dirt or grime. The use of anything stiffer or more abrasive may result in damage to the surface. Acid or alkali cleaners should not be used, as these will damage anodic films and may discolour coloured hardware.
- Where greasy deposits or hard to remove grime is present, the anodising may be cleaned with a soft cloth dipped in white spirit, turpentine, kerosene, or a mild liquid scourer, followed by wiping it with a dry rag. However, the cleaner must ensure none of these solvents come into contact with other parts of the system. All solvents must be kept from contact with the Santoprene glazing gasket materials (the "rubber" seal around the glass), as most solvents will damage them.
- It is essential to rinse anodised aluminium thoroughly with copious applications of clean water after cleaning, particularly where crevices are present, and then dry the glass to prevent water spots.

Regularly washing anodised hardware will ensure a long lasting product. In general, the following programme is recommended:

- Rural environments: every six months.
- Urban environments: every three months.
- Industrial and marine environments: every six months, as well as a monthly cold water wash.

For additional protection, especially in harsh environments, waxing with a good quality car wax after washing will assist in lifting and maintaining the appearance of your anodised hardware.

Damage to anodised surfaces may occur during building. Painters may accidentally splash paint on newly installed hardware, marring their appearance. The cleaner must act quickly and remove such splashes with a soft cloth moistened with water. Using water based paints allows the cleaner to clean with water - using solvents may put your hardware at risk.

LUGANO BALUSTRADE SYSTEM

CARE AND MAINTENANCE

CARE AND MAINTENANCE OF STAINLESS STEEL

Stainless steel is used for fittings and hardware for its strength, aesthetics, and its inherent high level of corrosion resistance. 2205 & 316 Grade Stainless Steel (also known as Marine Grade stainless) hardware is predominately used for external applications such as balustrades, pool fences, canopies and spider walls, but many of the entry door patch fittings and handles use 304 Grade. The design of our frameless glass systems and installation techniques are intended to make the systems as maintenance free as possible, but there are still maintenance procedures, associated with any exposed structure, that must not be overlooked. Stainless Steel is called so as it is less prone to staining - it is important to recognize that the material is not impervious to mild staining or even corrosion in some instances. The likelihood and severity of staining is a product of exposure to marine salts and other corrosive materials which can affect installations even 20km inbound from coastal areas. Pollen and other airborne matter can also contribute to corrosion so there are no areas of New Zealand where the topic can be ignored. The smoother the surface is, the less prone to discoloration.

Stainless steel may be discoloured from corrosion;

- If used in areas where rain does not wash the fittings.
- If it is exposed to a more aggressive environment than that for which the particular grade of steel is intended, e.g. highly polluted air, salt solutions or residues of cleaning agents containing chlorine.
- If it has a rough surface that enables a corrosive substance to adhere to.
- If the fittings are not cleaned after installation and have oil/acid from hands.
- If the design of the steel component has crevices and narrow gaps.
- If the surface is contaminated or damaged by grinding swarf or other iron particles from tools used in the installation work.
- If fasteners of ordinary steel or dissimilar metals are used for securing the hardware, or if the hardware comes into direct contact with adjacent components made of plain carbon steel in wet or humid conditions (galvanic corrosion).

Light corrosion is often referred to as "Tea Staining" as it is a brown surface discolouration which, although not affecting the material structurally, is none the less unsightly. The likelihood of this occurring can be greatly reduced through the use of 2205 or 316 Grade, regular cleaning and the use of protective coatings.

The following guidelines apply;

- a. Any stainless steel hardware should be cleaned after installation and before any glass is installed.
- b. Basic cleaning can be carried out using simple soap solutions or mild detergents applied with warm water (or proprietary cleaners) and a clean non-abrasive cloth. The solution should be thoroughly rinsed off with cold water, and wiped dry with a clean absorbent cloth. Isopropyl alcohol can also be used to clean finger and hand marks.
- c. Avoid bleach, as this can mark the metal surface, and avoid any abrasive applicators - especially a ferrous based cleaning pad, such as steel wool, as this can introduce contamination which reacts with the stainless steel and makes corrosion worse.
- d. Protective coatings are also available that can be easily applied to the hardware which can increase maintenance periods. Information regarding these products is available from OPUS. Stainless steel hardware should be cleaned again during the final clean and if possible after any surrounding building work is complete.
- e. If mild tea staining has already occurred, a plastic abrasive pad - "Scotchbrite" for example - can be used, again with warm water and a mild detergent / soap solution. When abrading however, it is important to only rub in one direction which is the same direction as any visible brushed finish. Take care to only rub the steel components, not the glass. A stainless steel rejuvenating paste can also be used and works well in combination with a scotchbrite pad. Information regarding these products is available from OPUS.
- f. When installation is complete the frequency of a regular cleaning regime will vary according to the installation design and level of exposure, especially in regard to proximity to the sea. In general cleaning should take place 3 to 4 times a year. Protective coatings can prolong the maintenance interval.

In summary:

- Cleaning should be thorough and at a frequency to suit the environment
- Wash with warm water and mild detergent or soap solution or use proprietary stainless steel cleaners.
- Rinse thoroughly with clean cold water and dry.
- Do not use harsh abrasive cleaners and especially no wire wool or similar ferrous scourers.
- If mild corrosion is present, then a mild abrasive detergent or rejuvenating paste can be used with a warm mild solution and fully rinsed with clean cold water.
- Heavier levels of staining can be removed with a light plastic abrading pad (such as Scotchbrite) rubbing must only be in one direction with that being as per any visible surface finish. This is best done with a rejuvenating paste □ Use protective coatings after cleaning

Be careful to ensure cleaners do not effect fibre or other fitting gasket materials

LUGANO BALUSTRADE SYSTEM

CARE AND MAINTENANCE

CARE AND MAINTENANCE OF HANDLES / MECHANISMS

Periodic maintenance of handles / mechanisms is essential, this includes locks, handles, hinges, levers, rollers, bolts, etc. This applies particularly to mechanisms with moving parts that require lubrication. Before doing any cleaning or maintenance, you must establish exactly what hardware has been installed and how it has been constructed.

Lubrication for Mechanisms;

Mechanisms include hinges, cylinders, locks, rollers and fasteners. You can keep these mechanisms in good working order through regular cleaning and lubrication.

- OPUS recommend a Teflon-based lubricate or please refer to the manufacturer's care instructions. A soft bristle brush can be used on exposed parts. Apply Teflon-based lubricate to the moving parts - you don't need to use very much. This will limit corrosion of the exposed metals. OPUS advise to do this once every two months. If, however your hardware is near the sea or exposed to salt air, we recommend you lubricate the components once every month.

Electrical Entrance Systems;

- Do not let the product get wet.
- Keep clear of debris and general dirt.
- Wipe the keypads or swipe the device with a clean damp cloth.
- Do NOT use solvents.
- Replacement batteries will be required.

SCRATCHES AND METAL SCRAPERS

Scratches can occur from hard pointed objects or poor handling, but most often occurs from the careless removal of foreign matter from the glass surface. Mortar splatter and paint are common offenders and efforts to remove after hardening almost always lead to surface damage.

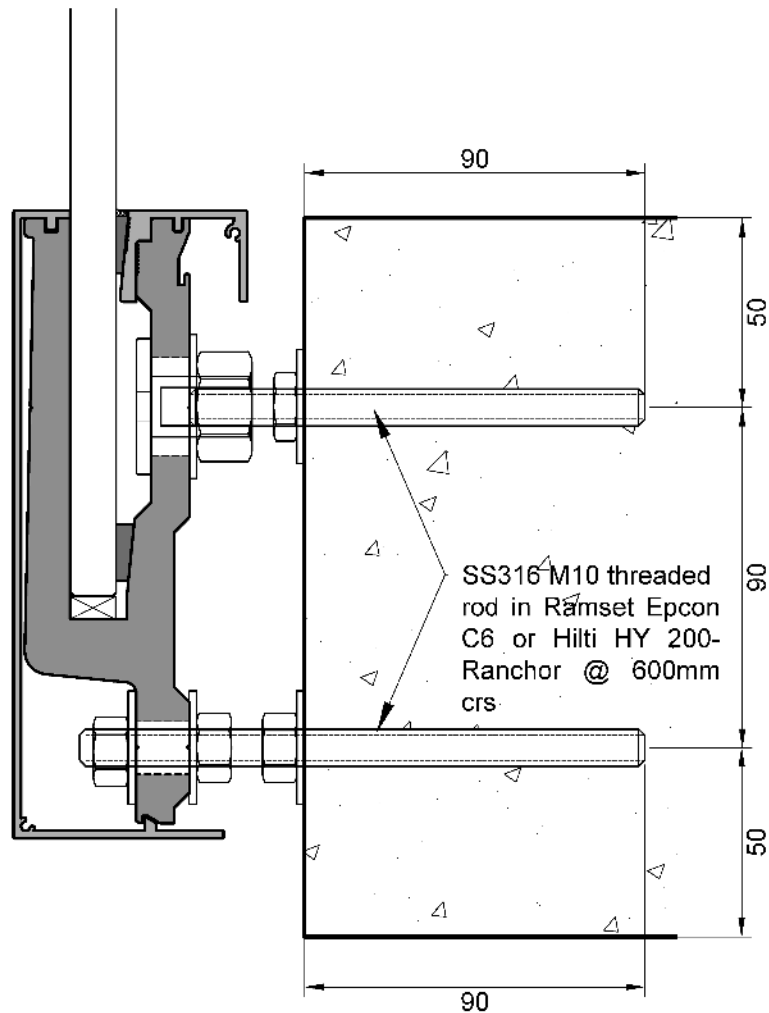
- It is essential that the foreign materials are removed before they harden. Better still, if construction work continues after glazing, that the glazed areas are protected by adhesive plastic films or suitable covers.
- One of the common mistakes made by non-glass trades people, including glass cleaning contractors, is the use of razor blades or other metal scrapers on a large portion of the glass surface. Using large blades to scrape a window clean carries considerable risk of causing damage to the glass.
- The glass industry, fabricators, distributors and installers neither condones nor recommends any scraping of glass surfaces with metal blades or knives. Such scraping usually permanently damages or scratches the glass surfaces.
- When paint or other construction materials cannot be removed with normal cleaning procedures, a new 25mm razor blade may have to be used. The razor blade should be used on small spots only. Cleaning should be done in one direction only. Never scrape in a back and forth motion as this could trap particles under the blade that could scratch the glass. Blades or scrapers can dislodge "pickup" on toughened and heat strengthened glass. There are fine particles of glass that are fused on to the surface during toughening. Once dislodged they can scratch the glass.

ONSITE CONSTRUCTION / INSTALLATION PROTECTION

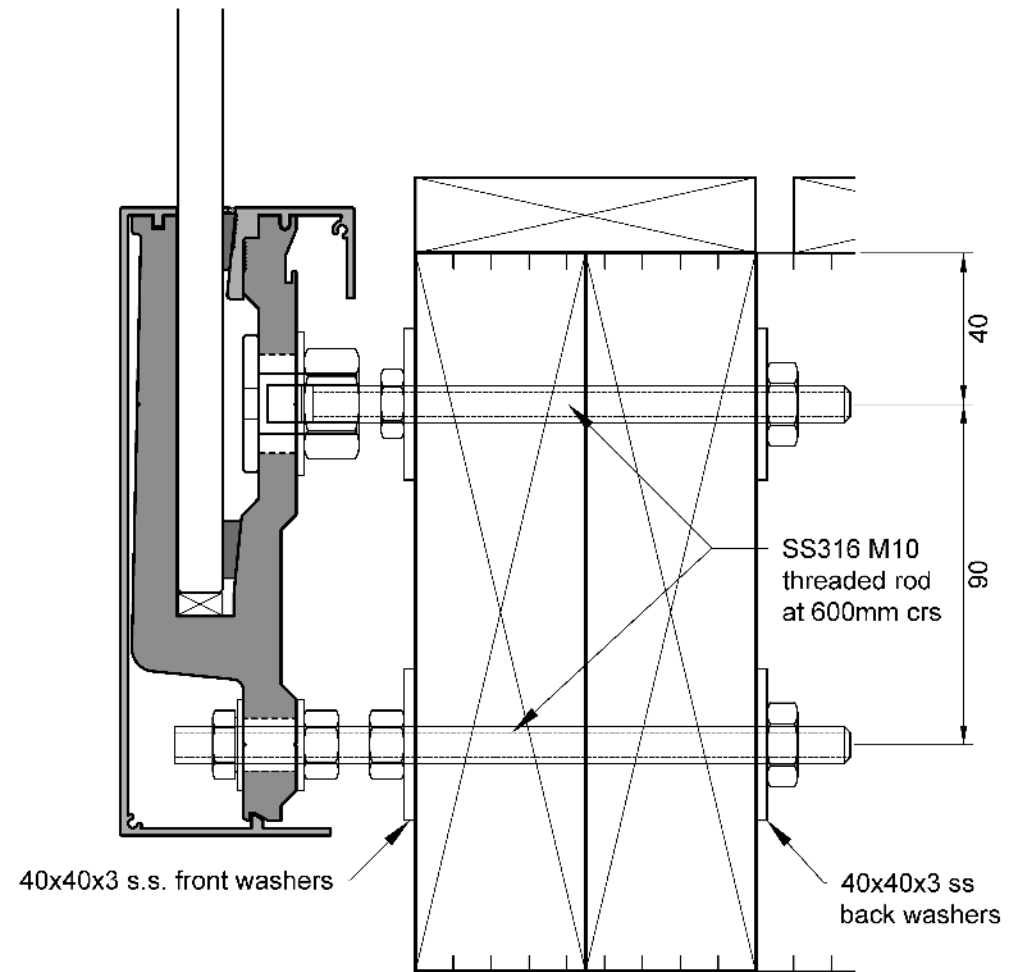
Hardware Protection during Installation:

- All the activity on a construction site means that your hardware items may get knocked or scratched, splattered with mortar, plaster, textured coating or paint during the later stages of construction.
- Please ensure that all hardware articles are masked or covered at this time. It is far easier to prevent accidents than to try and correct them. Should your hardware receive mortar or paint splashes see that these are removed before cure and follow the instructions outlined above.

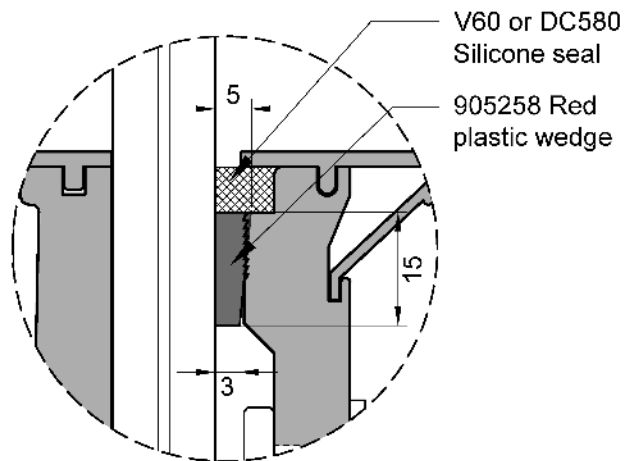
For further information, please refer to the PS1 or OPUS instructions and guidelines on our website.



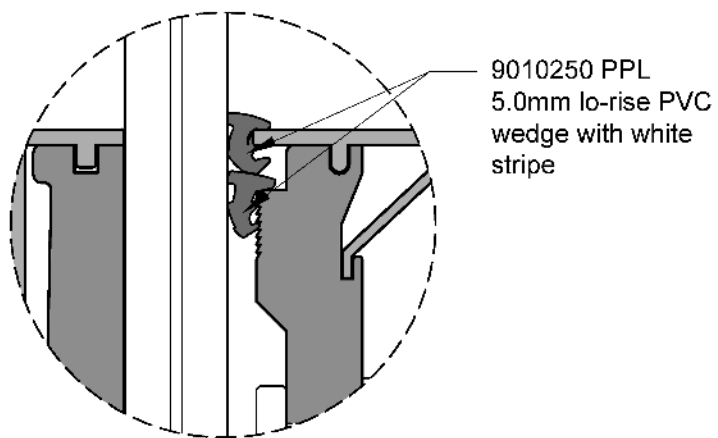
DETAIL A



DETAIL B



OPTION 1



OPTION 2

